

Method Description: Robust AI-Generated Image Detection via SigLIP2-Giant and Perturbation-Aware Training

1. Overview

Our approach proposes a simple yet highly effective framework for detecting AI-generated images in the wild. We leverage a pre-trained Vision-Language model backbone combined with a streamlined linear classification head. To ensure the model remains reliable under severe real-world distribution shifts, we explicitly integrate the official image distortion pipeline into our training phase, forcing the network to learn perturbation-invariant representations.

2. Model Architecture

The architecture consists of two main components:

Backbone Feature Extractor: We experimented with various scales of the SigLIP2 model family and empirically found that the largest variant, siglip2-giant-opt-patch16-384, achieved the highest performance on the leaderboard. To adapt this foundational model for downstream binary classification, we extract the sequence of patch embeddings from its final hidden layer.

Feature Aggregation & Classification Head: While we explored complex feature aggregation strategies—including CLS token extraction, multi-head attention pooling, and concatenating multiple layers—our results demonstrated that Global Average Pooling (Mean Pooling) across all patch tokens of the final layer yields the most robust and stable performance. The aggregated feature vector is then directly fed into a single Linear layer to output the final real/fake logits.

3. Robust Data Augmentation

The core of our robustness strategy lies in the data preprocessing pipeline. Real-world images are routinely subjected to various transformations. To simulate this environment and prevent overfitting to clean data, we apply a specific composition of augmentations during training:

Spatial Transforms ("Squish" Strategy): Instead of using random resized cropping which might discard crucial localized artifacts, we employ a "squish" transformation strategy. All input images are directly resized to 384×384 pixels (ignoring the original aspect ratio) to preserve the entire global image content, followed by Random Horizontal Flipping.

Competition-Specific Distortions: We wrapped the official `distort_images` function into our PyTorch transform pipeline. During our best training run, we set the distortion probability to **100%** (`distortion_prob=1.0`). This means every single training

image undergoes up to 3 random post-processing distortions at various severity levels (`num_levels=5`). This aggressive, explicit augmentation acts as a strong regularizer and is the key driver behind our high Robust ROC AUC scores.

4. Training Details

The model was trained using the Hugging Face `Accelerate` library for distributed multi-GPU training. The specific hyperparameters for our best-performing submission are as follows:

Optimization: We optimized the network using AdamW with a batch size of 24 per GPU. To preserve the foundational knowledge of the giant backbone while quickly adapting the new head, we used differential learning rates: $1e-5$ for the pre-trained SigLIP2 backbone and $3e-4$ for the classification head.

Learning Rate Schedule: A Cosine Annealing learning rate scheduler with a linear warmup phase (10% of total training steps) was applied over a total of 5 epochs.

Hardware & Acceleration: Mixed precision was enabled to accelerate training and reduce memory footprint while maintaining numerical stability.